

The problem could instead be treated as three independent sources, all of which are true ray fields and thus have no ambiguity in the direction of incidence. This can be achieved by setting the value of `ray_output` to 'all' instead of 'none'. A more accurate description is then obtained, and it is possible to observe differences between the two approaches, although at a very low field level. For most practical calculations it will be sufficient to use the default value of `ray_output` ('none').

The above example describes the concept by looking at the GTD field from a single reflector. However, the same approach is applicable to e.g. an array where, by setting the value of `ray_output` to 'all', the contribution from each individual element of the array can be calculated separately rather than calculating the composite field. In a similar manner a PO current distribution can be viewed as an array of current elements, each of which acts as a point source.

There are two more options for the value of `ray_output`. These are 'spherical' and 'plane'. The effect of `ray_output` = 'spherical' is that the direction of propagation of the field at the field point is assumed to originate from the origin of a coordinate system specified by the user. This makes it possible to use a PO current distribution as a source for a subsequent GTD calculation, which is not allowed when `ray_output` = 'none'. It could be useful and speed up computations in cases where the behaviour of the source field is known in advance, for example the current distribution on a hyperboloidal subreflector illuminated by a point-source feed at the focus.

When `ray_output` = 'plane' the direction of propagation of the field at the field point is assumed to be parallel to the z -axis of a coordinate system specified by the user.

It is difficult to say in general which method should be used, because it depends very much in the problem under consideration. If computing time is of no concern then the `ray_output` = 'all' should be used since this is the most accurate approach to the problem. However, in very many cases it will be a most reasonable assumption to use the direction of Poynting's vector (`ray_output` = 'none'), and thereby save computing time. It is left to the user of GRASP to make the final and appropriate decision, based on the actual geometry of interest.

4.5 Gaussian Beam Theory

The Gaussian feed concept provides a good model of the feed's far field as well as its near field, which is important e.g. when a reflector is in the near-field of its feed horn. Therefore, the internal spherical wave expansion calculation is not needed for this model.

4.5.1 The source model

The Gaussian feed is a point source radiating a tapered beam. This is obtained by positioning the source in complex space. The field of the source satisfies Maxwell's equations in the near field as well as in the far field.

The point source is a z -directed Huygens source, which consists of an electrical x -directed short dipole (a Hertzian dipole) and a y -directed short mag-

netic dipole, see Figure 4-76. The choice of the Huygens source entails the added advantage that in any direction its polarisation is co-polar to the linear x -polarisation, as defined according to Ludwig's 3rd definition, cf. Section 4.1.5.1.

The complete field expressions for the electric and magnetic short dipoles are well known (e.g. [Stratton, 1941] or [Hansen, 1988]). Here we will only give the expression for the Huygens source at large distances r ,

$$\mathbf{E}(r, \theta, \phi) = E_0 \frac{e^{-jk r}}{r} (1 + \cos \theta) (\cos \phi \hat{\boldsymbol{\theta}} - \sin \phi \hat{\boldsymbol{\phi}}) \quad (4.164)$$

where E_0 is a constant.

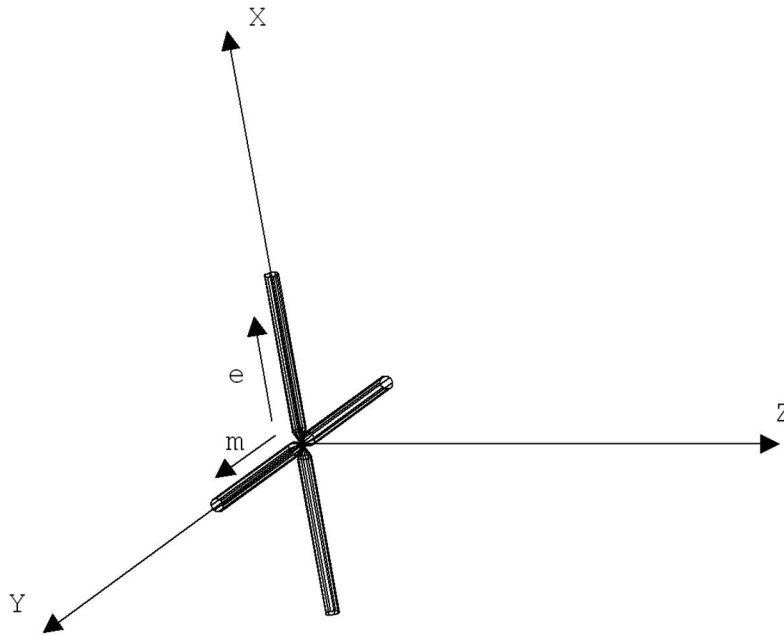


Figure 4-76 The Huygens source consisting of an electric and a magnetic dipole.

When a source is located at a point with the complex position $(x, y, z) = (0, 0, -jb)$ it will generate a Gaussian beam in the positive z -direction. The distance b is denoted the imaginary displacement. We will examine this by considering the field from a scalar point source

$$u = \frac{e^{-jkR}}{R} \quad (4.165)$$

where R is the distance from the source to the field point. The function u represents an exact solution to Helmholtz' equation for complex source points and/or field points and it has the same R -dependence as appears in (4.164). It is therefore well suited for investigating the properties of the Gaussian beam.

The distance R is in general a complex quantity. Obviously, only field points in real space are of practical interest and we will therefore investigate (4.165) in more detail for real field points $P(x, y, z)$.

The distance from the source to the field point is then

$$R = \sqrt{x^2 + y^2 + z^2 - b^2 + j2bz}. \quad (4.166)$$

The field is seen to be rotationally symmetric around the z -axis and for later use we introduce

$$\rho^2 = x^2 + y^2. \quad (4.167)$$

If we express P in spherical $r\theta\phi$ -coordinates we have

$$z = r \cos \theta \quad (4.168)$$

and if P is in the far field, $b \ll r = \sqrt{x^2 + y^2 + z^2}$, we find

$$\begin{aligned} R &\cong \sqrt{r^2 + j2br \cos \theta} \\ &\cong r + jb \cos \theta \end{aligned} \quad (4.169)$$

which inserted into (4.165) gives the expression for the field at far field points

$$u = e^{kb \cos \theta} \frac{e^{-jkr}}{r}. \quad (4.170)$$

This expression shows that the complex point source radiates a beam in the positive z -direction and the beam width is controlled by the parameter b .

The field from a scalar point source, (4.165), modifies for far field points to (4.170) when the source is moved into the complex space. It can be shown, that (4.164) in the same way modifies to

$$\mathbf{E}(r, \theta, \phi) = N_f \frac{e^{-jkr}}{kr} e^{kb \cos \theta} (1 + \cos \theta) (\cos \phi \hat{\boldsymbol{\theta}} - \sin \phi \hat{\boldsymbol{\phi}}) \quad (4.171)$$

for far field points when the Huygens source is moved to the complex position $(x, y, z) = (0, 0, -jb)$. The normalisation constant, which normalises the field to dBi is denoted N_f

From the above expression it is seen that the beam width is controlled by the parameter b . The far-field taper at a given angle θ can be determined from (4.171). If the taper is specified to A dB ($A < 0$) at the angle θ , this is obtained for

$$b = \frac{20 \log_{10}((1 + \cos \theta)/2) - A}{20k(1 - \cos \theta) \log_{10} e} \quad (4.172)$$

and A expressed by b is

$$A = 20 \log_{10}((1 + \cos \theta)/2) - 20bk(1 - \cos \theta) \log_{10} e. \quad (4.173)$$

4.5.1.0.1 Comparison to an accurate feed model

It is interesting now to compare the field from a realistic feed horn with that of the complex source. Figure 4-77 shows near-field patterns as well as the far-field pattern of a conical corrugated feed horn under balanced conditions. The aperture diameter is 5λ and the half flare angle is 9° . The field is plotted on spheres with radius r_m and centre at the far-field phase centre. The similar plots of the field from a complex source point are shown in Figure 4-78, the quantity b being determined such that the far-field taper at $\theta = 15^\circ$ is the same for the two cases, i.e. 12.6 dB giving a b of 6.7λ .

Comparing Figure 4-77 and Figure 4-78 it is seen that within the main beam region down to about 10-15 dB below the peak (the usual region used for illumination of the subreflector in a reflector antenna system) the complex source point model yields results that are completely satisfactory for practical applications.

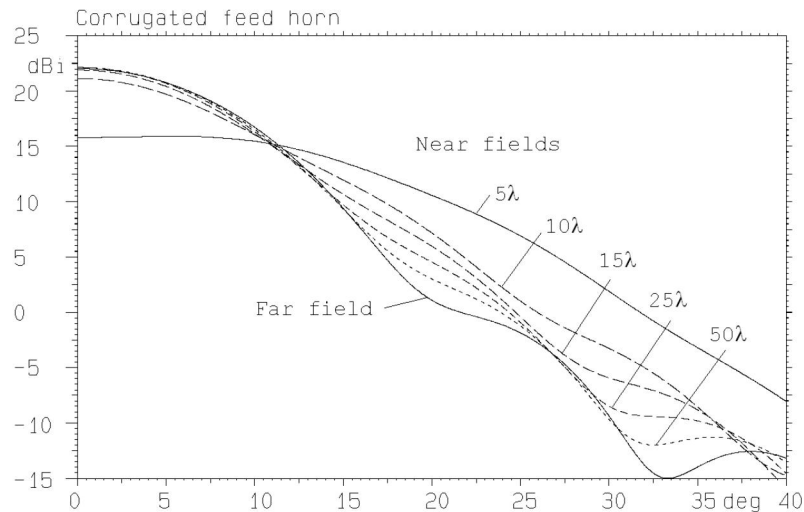


Figure 4-77 Fields of conical corrugated feed horn at different near field sphere radii.

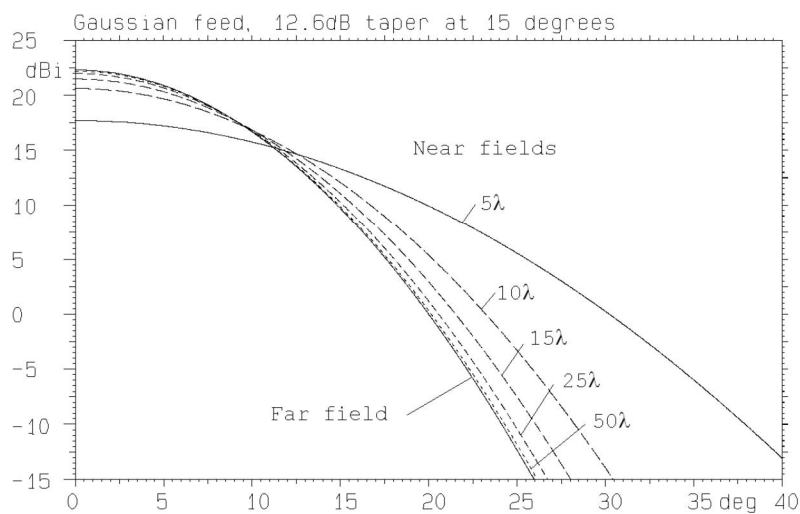


Figure 4-78 Fields of complex source point, corresponding to corrugated feed horn above.

4.5.1.0.2 Near-field properties of the complex point source

Let us again consider the field from the simple scalar source, (4.165). For near-field points in the plane $z = 0$ we have

$$R = \begin{cases} \sqrt{\rho^2 - b^2}, & \text{for } \rho > b \\ j\sqrt{b^2 - \rho^2}, & \text{for } \rho < b. \end{cases} \quad (4.174)$$

As practically all power is carried within $\rho < b$ we apply the latter expression and obtain for the near field

$$u = \frac{1}{j\sqrt{b^2 - \rho^2}} e^{k\sqrt{b^2 - \rho^2}}. \quad (4.175)$$

If we normalise this field to unity for $\rho = 0$ the normalised field may be written as

$$u_n = \frac{1}{\sqrt{1 - \frac{\rho^2}{b^2}}} e^{kb\left(\sqrt{1 - \frac{\rho^2}{b^2}} - 1\right)}. \quad (4.176)$$

Figure 4-79 shows a plot of u_n as a function of $\frac{\rho}{\lambda}$ for $b = 2\lambda$, 5λ and 10λ . The expressions for u and u_n become singular at $\rho = b$ but ρ must be very close to b before the singular factor is important. For example, for $\rho = 0.99999b$ the singular factor has increased to only 220 while the exponential term is $3.5 \cdot 10^{-6}$, even for b as low as 2λ . The product of the two terms is therefore still small and the effect is invisible in the figure.

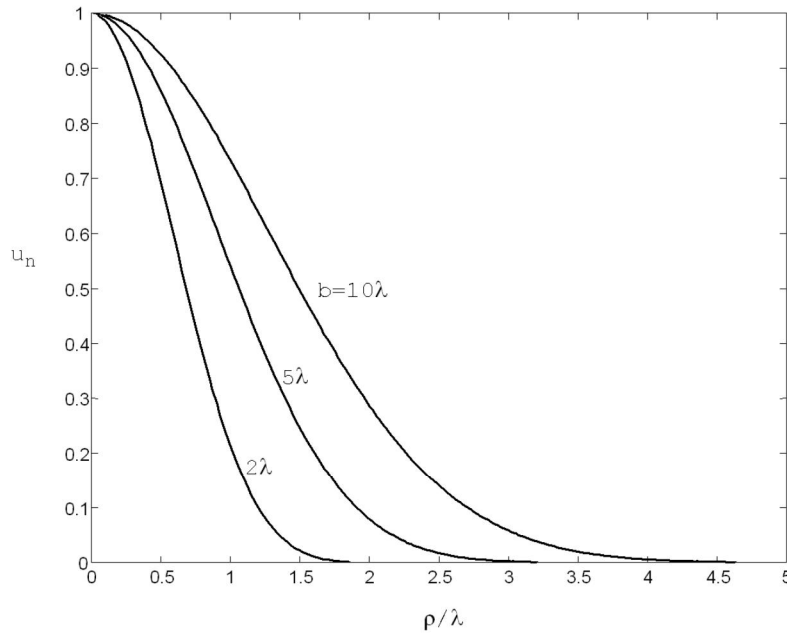


Figure 4-79 Normalised field u_n in plane $z = 0$ for a point source located at $(0, 0, -jb)$, $b = 2\lambda$, 5λ and 10λ .

4.5.2 Gaussian beams

We will now demonstrate that the near field of the complex point source agrees with the usual Gaussian beam formulas. In the so-called paraxial region (near the axis), $z^2 + b^2 \gg \rho^2$, we have by definition (4.166)

$$\begin{aligned} R &= \sqrt{\rho^2 + (z + jb)^2} = (z + jb) \sqrt{1 + \frac{\rho^2}{(z + jb)^2}} \\ &\cong z + jb + \frac{1}{2} \cdot \frac{\rho^2(z - jb)}{z^2 + b^2}, \end{aligned} \quad (4.177)$$

which gives

$$\frac{e^{-jkR}}{R} = -\frac{j}{b} e^{kb} \frac{w_0}{w} \exp\left(-\frac{\rho^2}{w^2} - j\frac{k\rho^2}{2R_c} - j(kz - \Phi)\right), \quad (4.178)$$

where

$$w^2(z) = w_0^2 \left(1 + \left(\frac{z}{b}\right)^2\right), \quad w_0^2 = \frac{2b}{k} \quad (4.179)$$

$$R_c(z) = z \left(1 + \left(\frac{b}{z}\right)^2\right) \quad (4.180)$$

$$\Phi(z) = \arctan \frac{z}{b}. \quad (4.181)$$

Here $w(z)$ is the Gaussian beam radius, which is the lateral distance where the field is decayed by $1/e$. It varies along the beam and has its minimum value w_0 in the beam waist. The variation of the beam radius along z (the beam axis) is given by (4.179), where z is the distance to the waist and k the wave number. The cross-sectional amplitude distribution of the beam for fixed z is

$$\exp(-\rho^2/w^2), \quad (4.182)$$

where ρ is the distance from the beam axis.

The beam phase front has a radius of curvature R_c which varies with distance as given by (4.180).

It is noted that the radius of curvature, R_c , is the reciprocal of the curvature. A typical beam is illustrated in Figure 4-80. The two points, F_1 and F_2 , are located the distance b from the beam axis. It can be shown that the surfaces of constant phase are ellipsoids, which in the cut of the figure are ellipses with focal points F_1 and F_2 . The ellipsoids are obtained by rotating the figure around the beam axis. The orthogonal curves in the figure (hyperbolas with the same foci) represent surfaces of constant amplitude relative to the value on the axis.

If w and R_c are given the relations (4.179) and (4.180) may be inverted to give

$$w_0 = \frac{w}{\sqrt{1 + \left(\frac{kw^2}{2R_c}\right)^2}}, \quad z = \frac{R_c}{1 + \left(\frac{2R_c}{kw^2}\right)^2}. \quad (4.183)$$

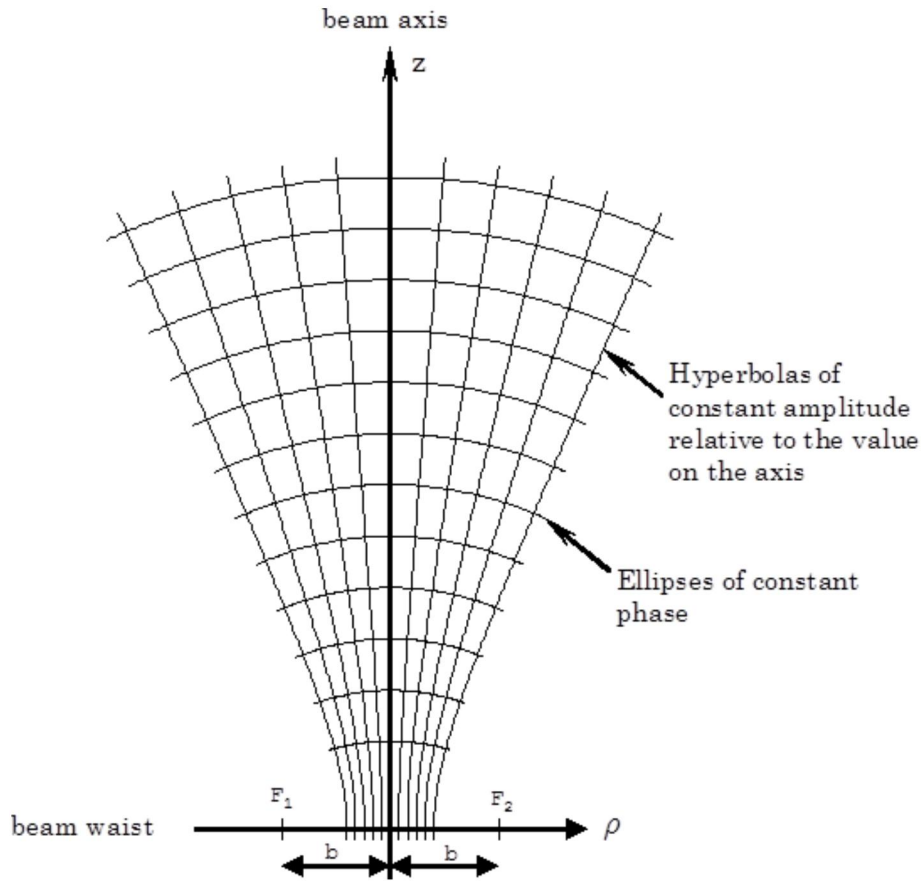


Figure 4-80 A typical Gaussian beam. The beam is rotationally symmetric around the axis.

The centre of curvature of the beam field is located b^2/z behind the waist. At the beam waist, $z = 0$, the radius of curvature of the wave front is infinite. For increasing z the radius of curvature decreases to its minimum value, $R_c = 2b$, for $z = b$. For large z the centre of curvature converges towards $z = 0$.

In most cases the Gaussian beam shall represent a feed which is characterised in terms of usual far-field data, such as the taper at a specific angle from bore-sight. The relationship between the beam waist radius w_0 and the taper A in dB at a given angle, θ is

$$w_0^2 = \frac{2}{k} \cdot \frac{20 \log_{10}((1 + \cos \theta)/2) - A}{20k(1 - \cos \theta) \log_{10} e}. \quad (4.184)$$

The above relation includes the fact that the Gaussian beam in GRASP is generated by locating a Huygens' source in complex space and the Huygens' source has a radiation pattern of $(1 + \cos \theta)$.

The beam radius and radius of curvature can conveniently be put together in a so-called complex beam parameter, q , defined by

$$\frac{1}{q} = \frac{1}{R_c} - \frac{j}{kw^2/2} \quad (4.185)$$

which gives

$$q = z + jb, \quad (4.186)$$

where z is the distance to the waist, counted negative before the waist and positive after the waist.

This parameter is transformed in a very simple manner through the system. A propagation through the distance L will change the beam parameter according to

$$q_{out} = q_{in} + L, \quad (4.187)$$

where q_{in} is the initial value of q and q_{out} is the output value of q . When the beam hits a conic mirror reflector or lens with f -value f (often denoted the focal length) it continues as a Gaussian beam with a new waist position and size according to

$$\frac{1}{q_{out}} = \frac{1}{q_{in}} - \frac{1}{f}. \quad (4.188)$$

The real part of this relation gives

$$\frac{1}{R_{out}} = \frac{1}{R_{in}} - \frac{1}{f}. \quad (4.189)$$

corresponding to the fact that a conventional point source at the distance R_{in} from the reflector will focus at a distance of R_{out} . If R_{out} is positive the field from the reflector will diverge as coming from a focus located R_{out} behind the reflector. If R_{out} is negative the field from the reflector will converge to a focus located $-R_{out}$ in front of the reflector.

If the point source is specifically located at one of the foci of the conic mirror reflector the equation gives the distance to the other focal point by means of the f -value f , which thus defines the geometry of the surface.

Another feature which can be derived from the equations is that the beam radius is the same just before and after reflection.

4.5.3 Definitions of a complex point source

A complex point source can be defined in two ways in GRASP:

- by specifying the taper A at the angle θ (far field specification) in which case the imaginary displacement b is calculated from (4.172), or
- by specifying the beam radius, w in (4.179) and the radius of curvature, R_c in (4.180) (near field specification). If the Gaussian beam is supposed to model a corrugated horn of flare angle θ_0 and aperture diameter D_h , see Figure 4-81, it can be shown that for small values of θ_0 the beam radius in the aperture is $w = 0.32D_h$ while the radius of curvature is equal to the slant length.

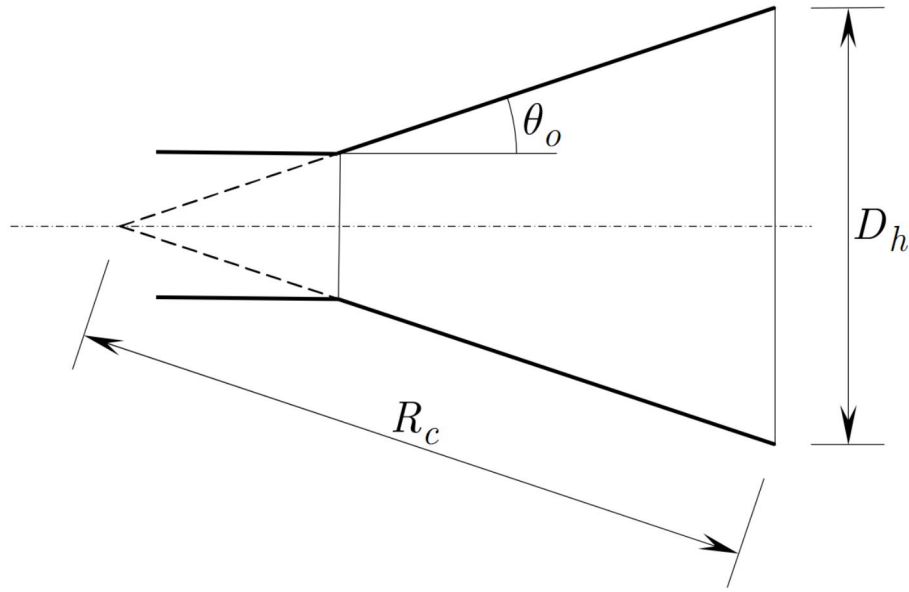


Figure 4-81 Horn parameters for the specification of an equivalent Gaussian beam.

For both options the polarisation can be specified by the user to be linear along the x - or the y -axis, or to be right- or left-hand circular.

4.6 Plane Wave Expansion

According to [Collin, 1969], an arbitrary electric field $\mathbf{E}$ can be expanded in a spectrum of plane waves of the form

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{f}(k_x, k_y) e^{-j\mathbf{k} \cdot \mathbf{r}} dk_x dk_y, \quad (4.190)$$

where $\mathbf{k}$ is the vector

$$\mathbf{k} = k_x \hat{\mathbf{x}} + k_y \hat{\mathbf{y}} + k_z \hat{\mathbf{z}} = k_x \hat{\mathbf{x}} + k_y \hat{\mathbf{y}} + \sqrt{k^2 - k_x^2 - k_y^2} \hat{\mathbf{z}} \quad (4.191)$$

where k is the wave number and $\mathbf{f}(k_x, k_y)$ is the amplitude density function, which is related to the far field by

$$\mathbf{E}_{far}(k_x, k_y) = j \frac{k k_z}{2\pi} \mathbf{f}(k_x, k_y). \quad (4.192)$$

The variables $k_x k_y k_z$ are related to the standard spherical angles θ and ϕ by

$$\begin{aligned} k_x &= k \sin \theta \cos \phi \\ k_y &= k \sin \theta \sin \phi \\ k_z &= k \cos \theta. \end{aligned} \quad (4.193)$$

In this way the near field (4.190) can be calculated from the far field (4.192) by numerical integration. The representation (4.190) of the near field is only